



AGH UNIVERSITY OF KRAKOW, POLAND
4-6TH OF SEPTEMBER 2026

2026 MARS YARD UPDATE REPORT

roverchallenge.eu



Change Record

Revision	Date	Location of Changes	Description of Changes
1.0	11.08.2026	Initial issue	Initial issue of the document.

Agenda

- Important Links
- Deliverables and Links
- Important Information
- Organisational Matters and Reminders
- Plan of the Venue
- Mars Yard
- Droning Sub-Task (Cage, Markers)
- Maintenance Task (Panel, Markers)
- Deep Sampling (Layers)



Important Links

- [Schedule](#) (this file will be updated so you need to often check it until you complete the last task)! It will be also shown on the screen in the team zone
- [Rules and other files](#)
- [MY Update Report and MY model](#)

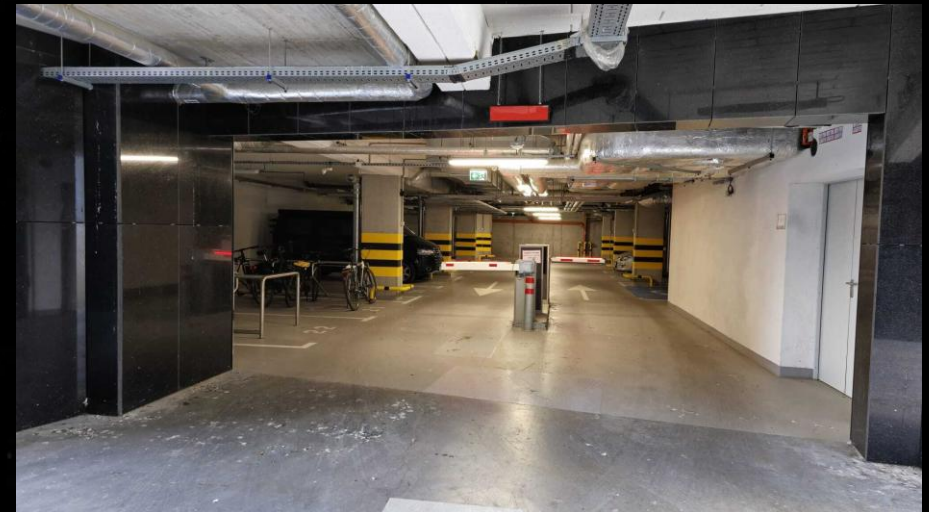
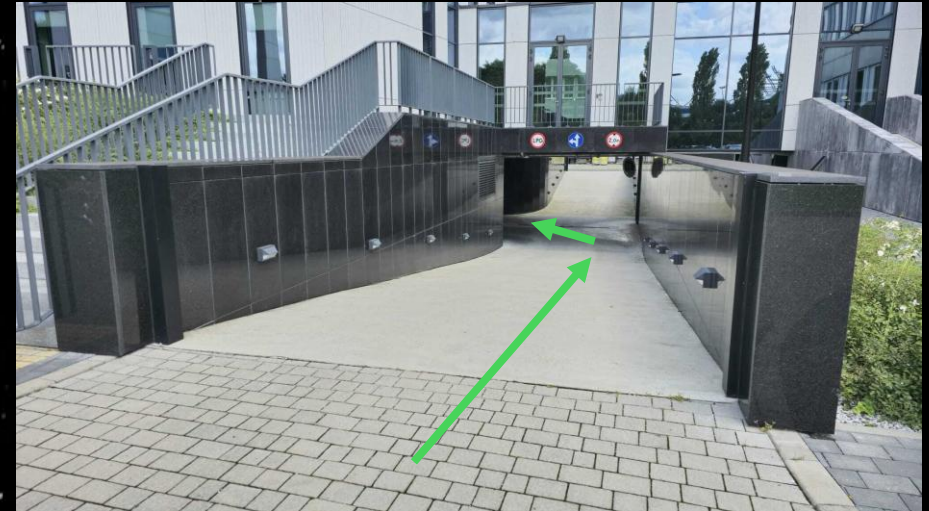


Important Links

No.	Event	Link	Deadline
4	Science Planning Report	https://forms.gle/kCqBdSf1HtePEQy28	23 rd August 2026 23:59 CEST / EOD
5	Droning Sub-Task Report	https://forms.gle/SczzpirtDypA4uQb9	2 nd September 2026 23:59 CEST / EOD
	Droning Sub-Task Photos	https://forms.gle/55xneJsci7uSQHfh8	up to 2.5 hours after the ride on the Mars Yard
6	Science Exploration / Final Report	https://forms.gle/SpvVA3vyrMACiEaD9	up to 2.5 hours after the ride on the Mars Yard
7	Science AstroBio / Final Report	https://forms.gle/adm2dFXY2QXMMd8HA	up to 2.5 hours after the ride on the Mars Yard
8	Presentation File (PDF preferred, no videos needed)	https://forms.gle/my8oZr4ymGZpLUfW8	5 th September 2026 23:59 CEST / EOD

Important Information

- Slack is our main communication platform for the event! Please install and turn on notifications. It will be the easiest way to reach the jury
- At the entrance to the **Team Zone** you'll find two locations: one for the **Team Coordinator (Konrad), Head of Jury (Marcin), Co-Head of Jury (Adam)** and the judges responsible for checking the **weight and RF of your rover (Milosz, Pawel, Grzegorz, Jan, Pawel)**. You will find also volunteers in the Team Zone. We're all there to help you, don't hesitate to contact us in case of any issue is encountered by your team
- **Remember about providing the Navigation-Traversal Planning Form to judges before your ride!**



Team Zone Photos ↑

Organisational Matters – Team Zone

Important Updates Regarding the Team Zone

- **Enhanced Safety & Work Comfort:** Due to safety concerns of third parties, we're implementing a 10-person limit per team for entry into the team zone. Each team will receive 10 IDs to manage as they see fit, granting access to their most crucial team members. This will significantly boost comfort and safety by reducing overcrowding and allowing for smoother movement.
- **Opening Hours:** The team area will be open 24 hours a day, but may be closed at night if the competitors are not interested in working at night. In this case, it will be opened in the morning approximately 30 minutes before the first weighting / RF check.
- **Team Size:** If a team has fewer than 10 members, they will receive IDs corresponding to their team size.

Organisational Matters – Team Zone

Important Updates Regarding the Team Zone

- Additional workspace - to accommodate for the new limits on the Team Zone area, we're introducing an additional area in a Studenckie Centrum Konstrukcyjne (SCK) building! This zone, equipped with desks, chairs, and power access, is designed for programmers, marketing specialists, and anyone who needs a calm place to work on their laptop. It's an ideal environment for focusing on coding, data analysis, or creating presentations.
- We're constantly working on introducing further conveniences and enhancements. We're open to your feedback and suggestions – your input is incredibly valuable to us as we strive to improve the event!



Organisational Matters – Reminders

Reminder #1:

We need to ask you a few basic questions about your trip to Cracow. Survey includes details related to car passes.

Deadline: Friday, 28th of August.

Moreover, please check shipment details in the newsletter sent you by Konrad.

See the link below:

<https://mailchi.mp/roverchallenge/erc-2026-shipping-address-10374761>

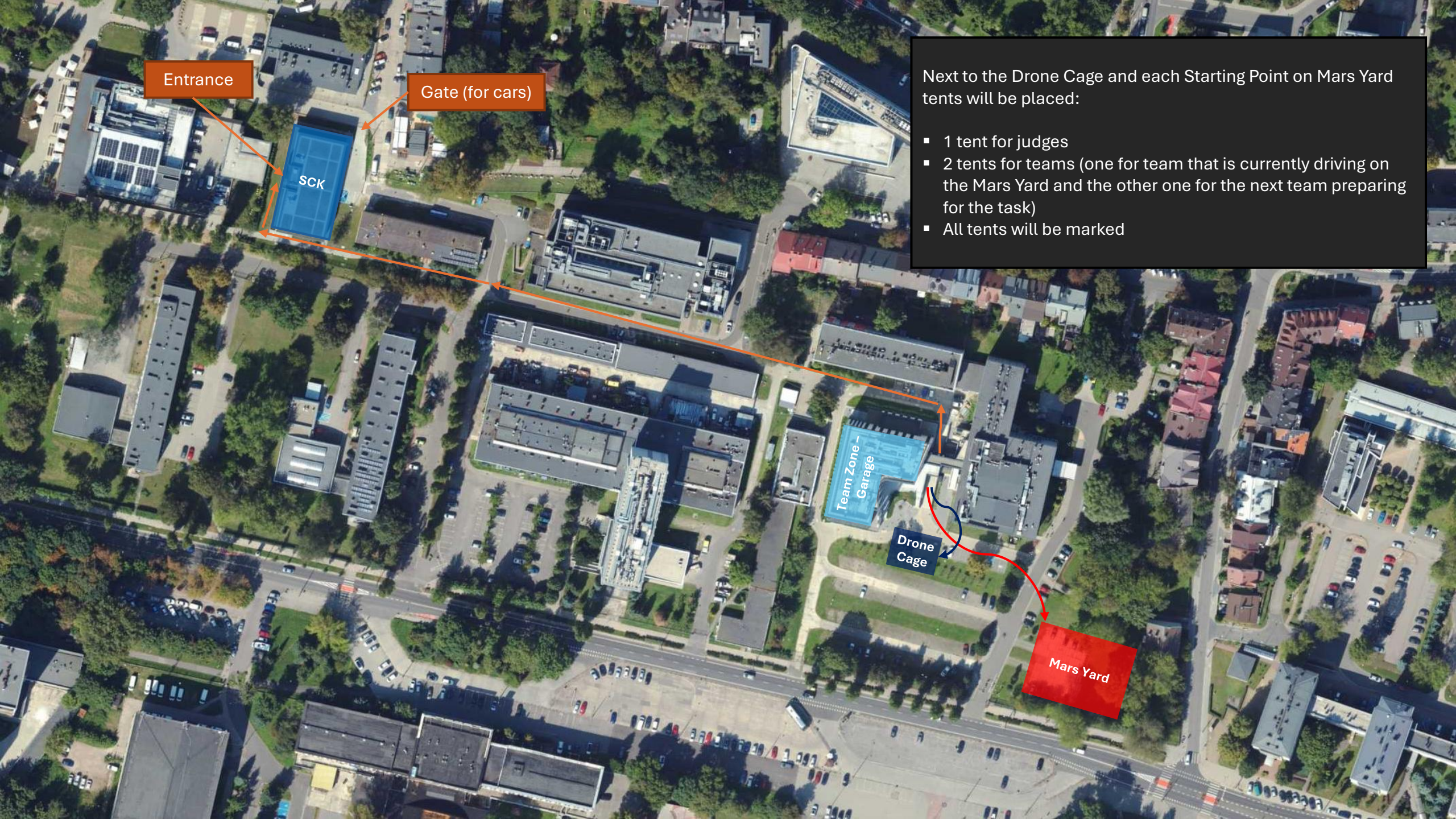
Reminder #2:

We need a signed consent to the processing of personal data from every team member (GDPR). Without a consent you won't receive a certificates after the event!

More details in the newsletter - see the link below:

<https://mailchi.mp/roverchallenge/erc-2025-onsite-rules-v2-10374098>





Entrance

Gate (for cars)

SCK

Team Zone -
Garage

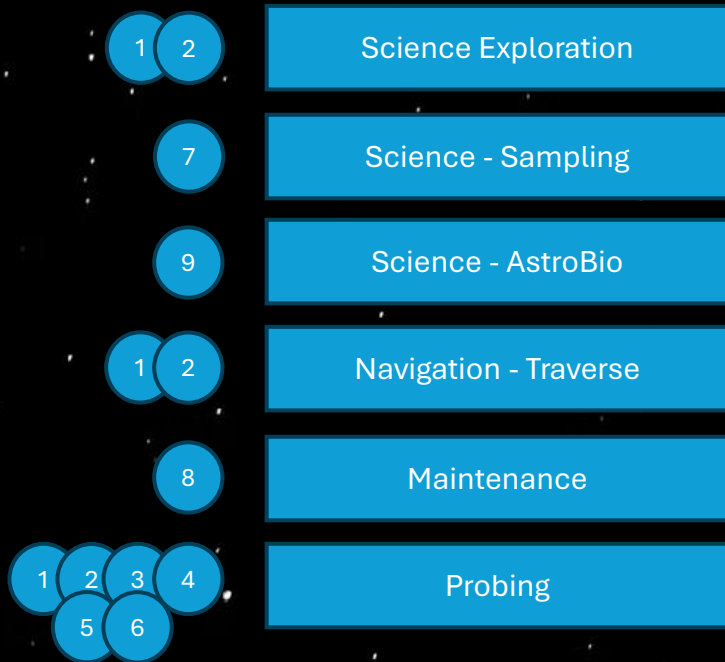
Drone
Cage

Mars Yard

Next to the Drone Cage and each Starting Point on Mars Yard tents will be placed:

- 1 tent for judges
- 2 tents for teams (one for team that is currently driving on the Mars Yard and the other one for the next team preparing for the task)
- All tents will be marked

Mars Yard Starting Points

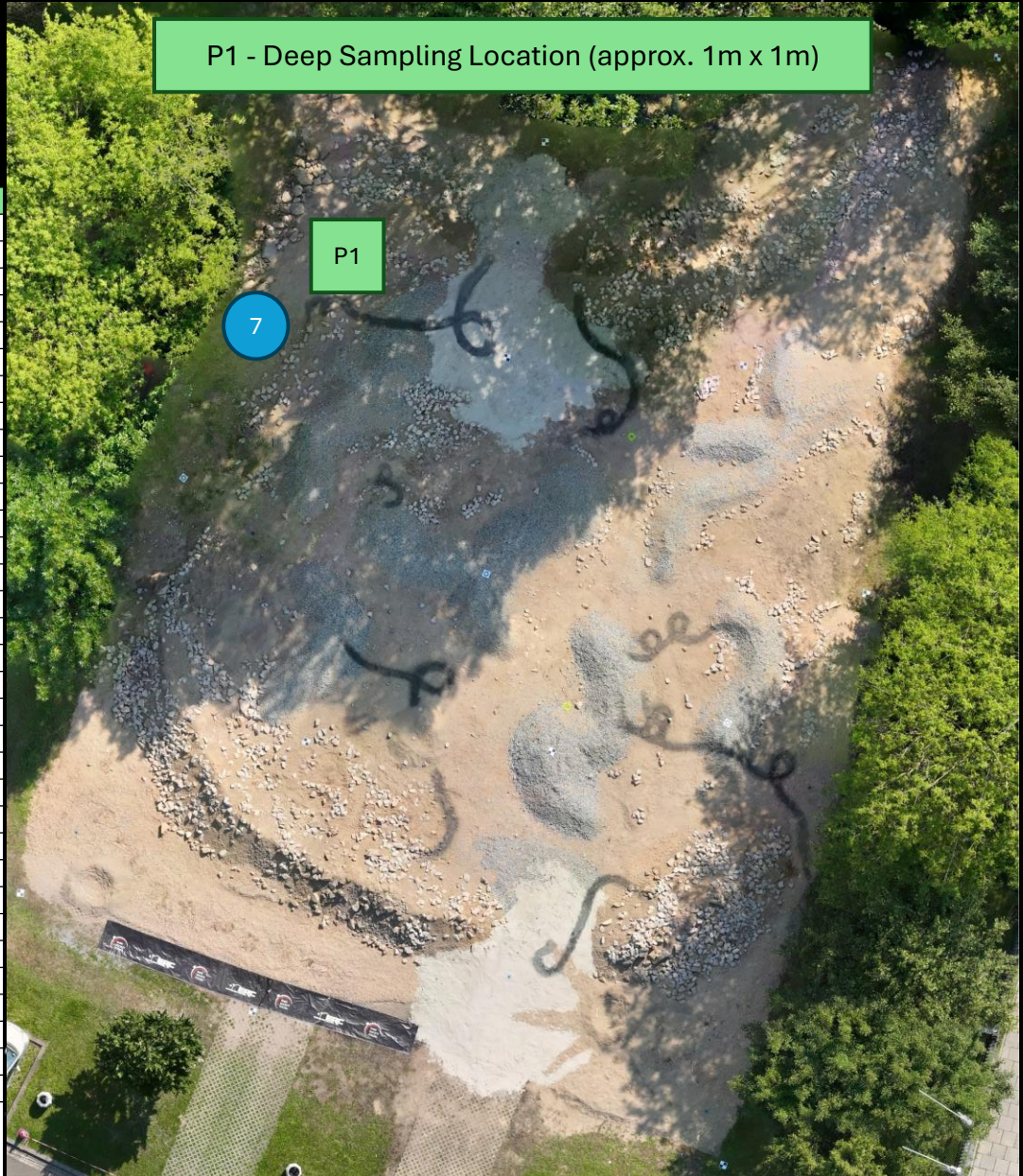


!!! ATTENTION !!!
Starting points have changed wrt to
the previous editions of the ERC.



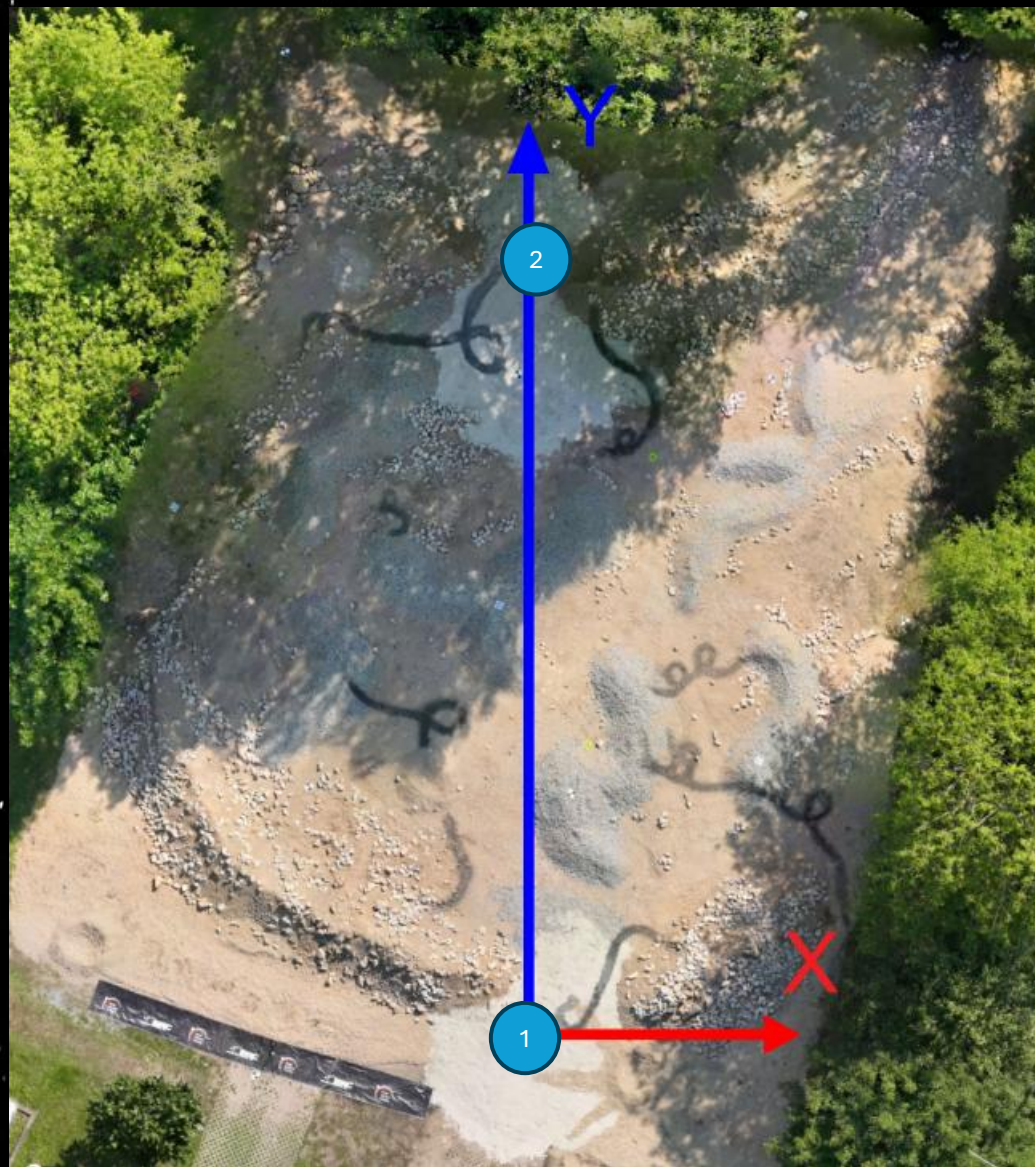
Mars Yard (All Points)

	Point	Y [m]	X [m]	H (local)
Starting Locations	S1	0,000	0,000	0,000
	S2	26,427	0,000	0,138
	S3	6,918	10,709	0,067
	S4	17,881	11,210	0,100
	S5	25,658	13,223	-0,035
	S6	15,824	-9,428	-0,302
	S7	22,831	-8,970	-0,180
	S8	4,991	-15,676	-0,280
	S9	-4,762	7,160	-0,095
Landmarks	L1	8,012	3,183	0,361
	L2	9,482	7,269	0,281
	L3	17,583	7,878	0,398
	L4	22,389	9,225	0,437
	L5	23,990	3,518	0,272
	L6	16,870	0,882	0,259
	L7	21,415	-3,944	0,104
	L8	16,334	-5,491	0,061
	L9	13,528	-7,695	-0,051
	L10	12,602	-1,610	0,290
	L11	9,721	-7,715	0,006
	L12	4,442	-4,311	-0,111
	L13	28,118	-5,720	-0,163
	L14	5,230	-11,438	-0,078
	L15	1,102	6,483	0,239
Navigation - Traverse Waypoints	W1	9,759	1,951	-0,053
	W2	19,499	4,170	0,105
	W3	23,149	7,013	0,205
	W4	18,665	-1,701	-0,156
	W5	19,034	-5,339	-0,259
	W6	8,197	-7,675	-0,179
	W7	7,682	-2,108	-0,163
	W8	6,790	5,939	-0,065
	W9	13,092	5,088	0,011
Deep Sampling Location	P1	25,901	-5,997	-0,160

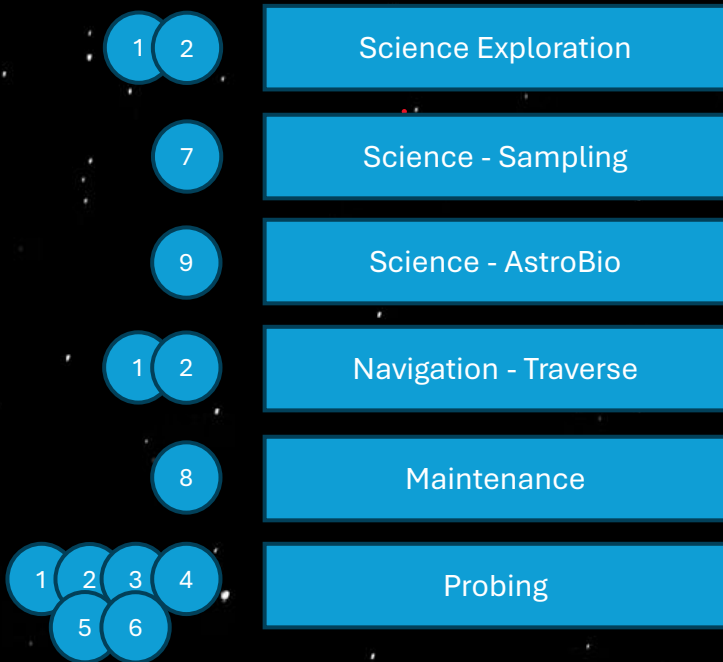


Coordinate System

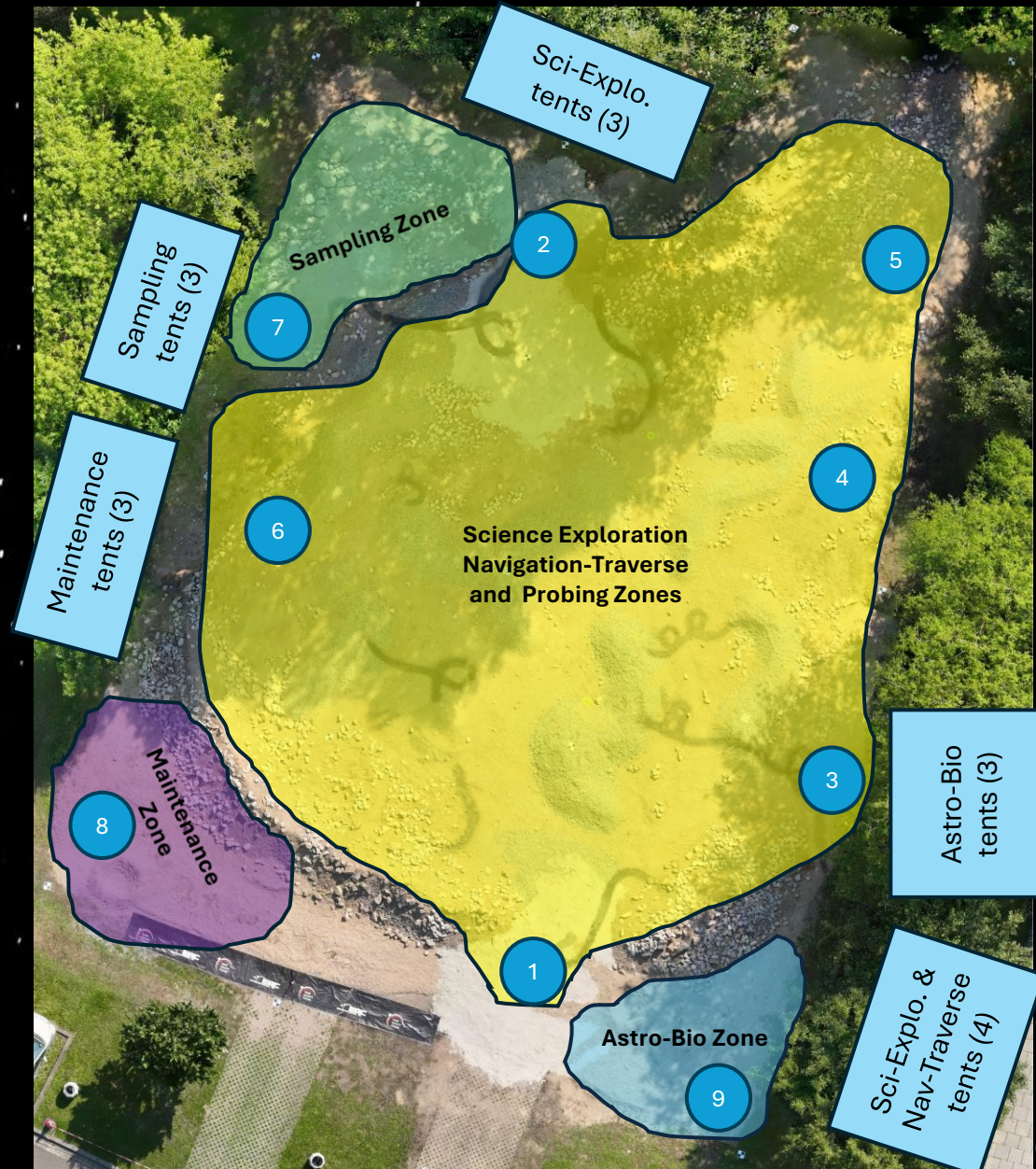
Points S1 and S2 are considered as calibration points (x coordinate = 0)



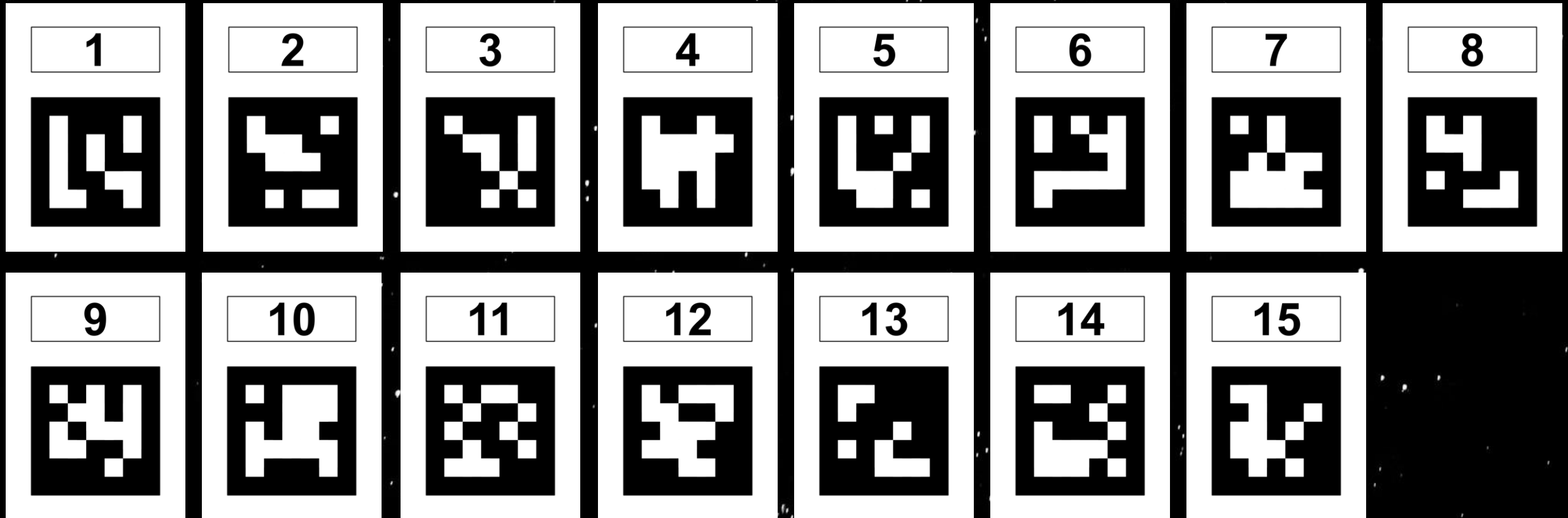
Mars Yard Zones



!!! ATTENTION !!!
Starting points have changed wrt to the previous editions of the ERC.



MY – Landmarks – ArUco Markers



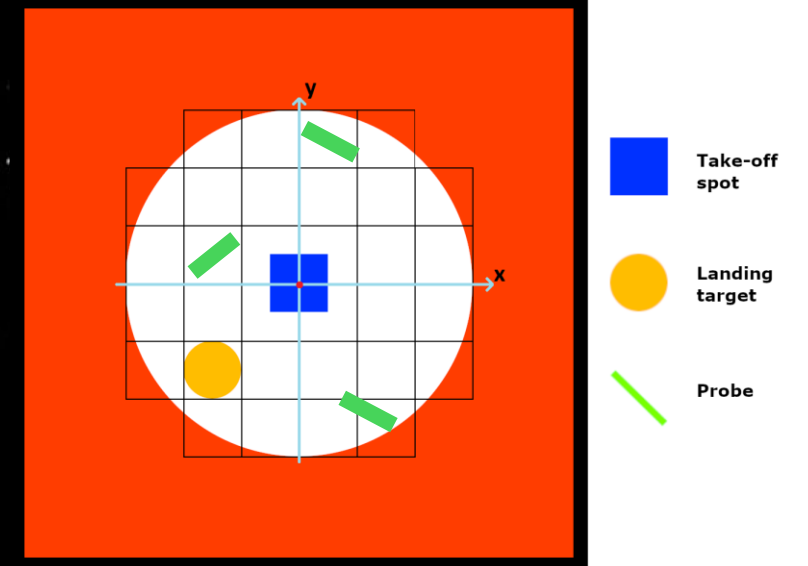
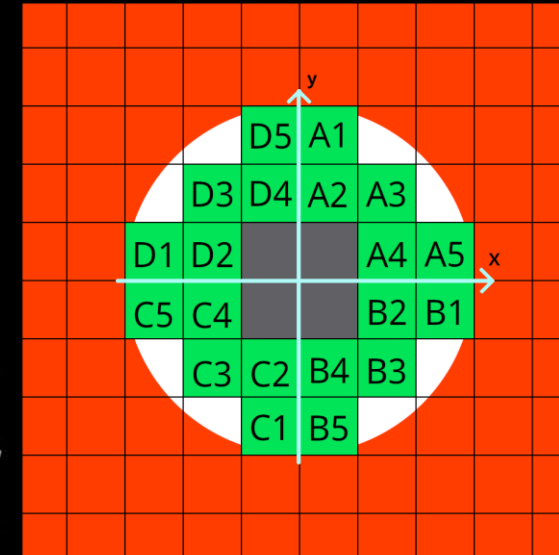
Additional details (e.g. related to landmark's size) are provided in the ERC Rules.
Markers from the 5x5 ArUco Library shall be considered (numbers 51-65).
See <https://chev.me/arucogen/>

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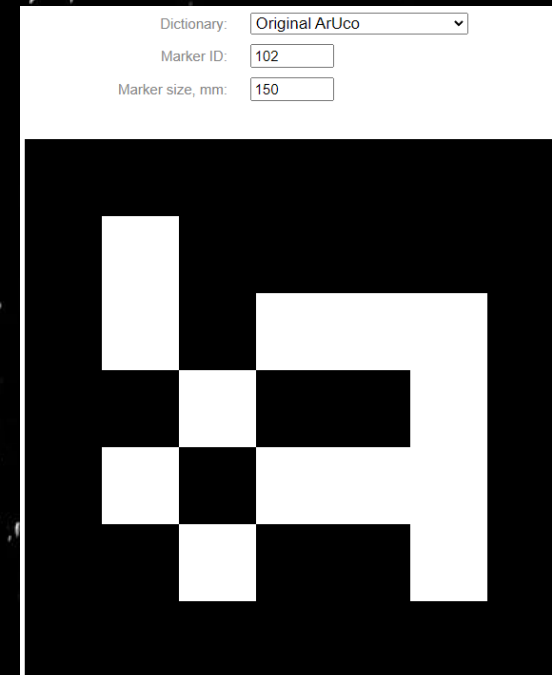
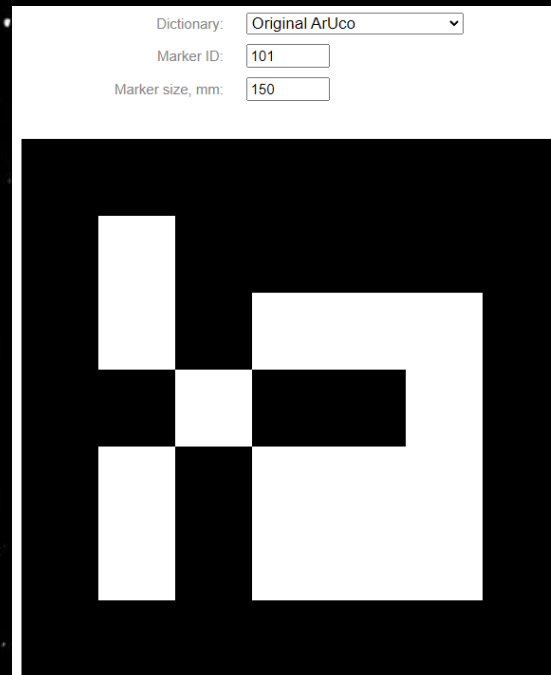
Droning Task - Cage

- Cage size: 10x10x4 m
- The effective area of competition is a disc with a radius of 3 m in the middle of the cage.
- The lift-off spot is a 1 x 1 m square in the middle of the cage.
- The landing target provided by the organizers is a disc with a radius of 0.5 m.
- Teams may also prepare their own landing targets attachable to their rover, but its size must be confined within a disc with a radius of 0.5 m.
- The lift-off spot and the landing target will be marked with an ArUco marker from the **ArUco original library**. The lift-off spot will use marker ID 101 and the landing target will use marker ID 102. The markers size will be 15 x 15 cm.
- The landing is considered as successful if any part of the drone is touching the landing target.
- The area of competition will be virtually divided into 1 x 1 m sectors. The probe detection system must point to locations of the probes using IDs of those sectors as shown on the diagram, below. As an example, sector A2 covers coordinates (0 m; 0.5 m) to (0.5 m; 1 m), D1 covers (-1.5 m; 0 m) to (-1 m; 0.5 m) etc.

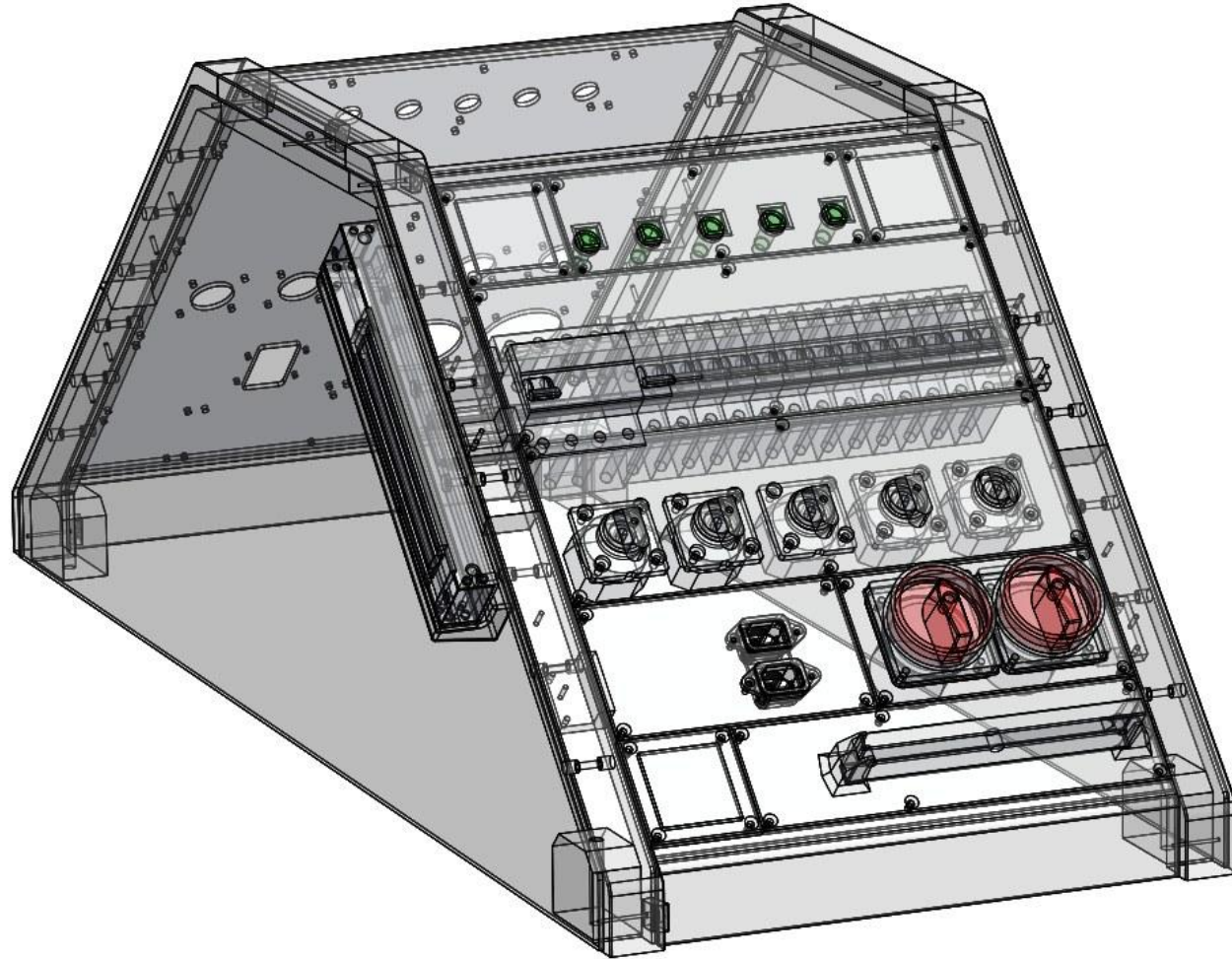


Droning Task - Markers

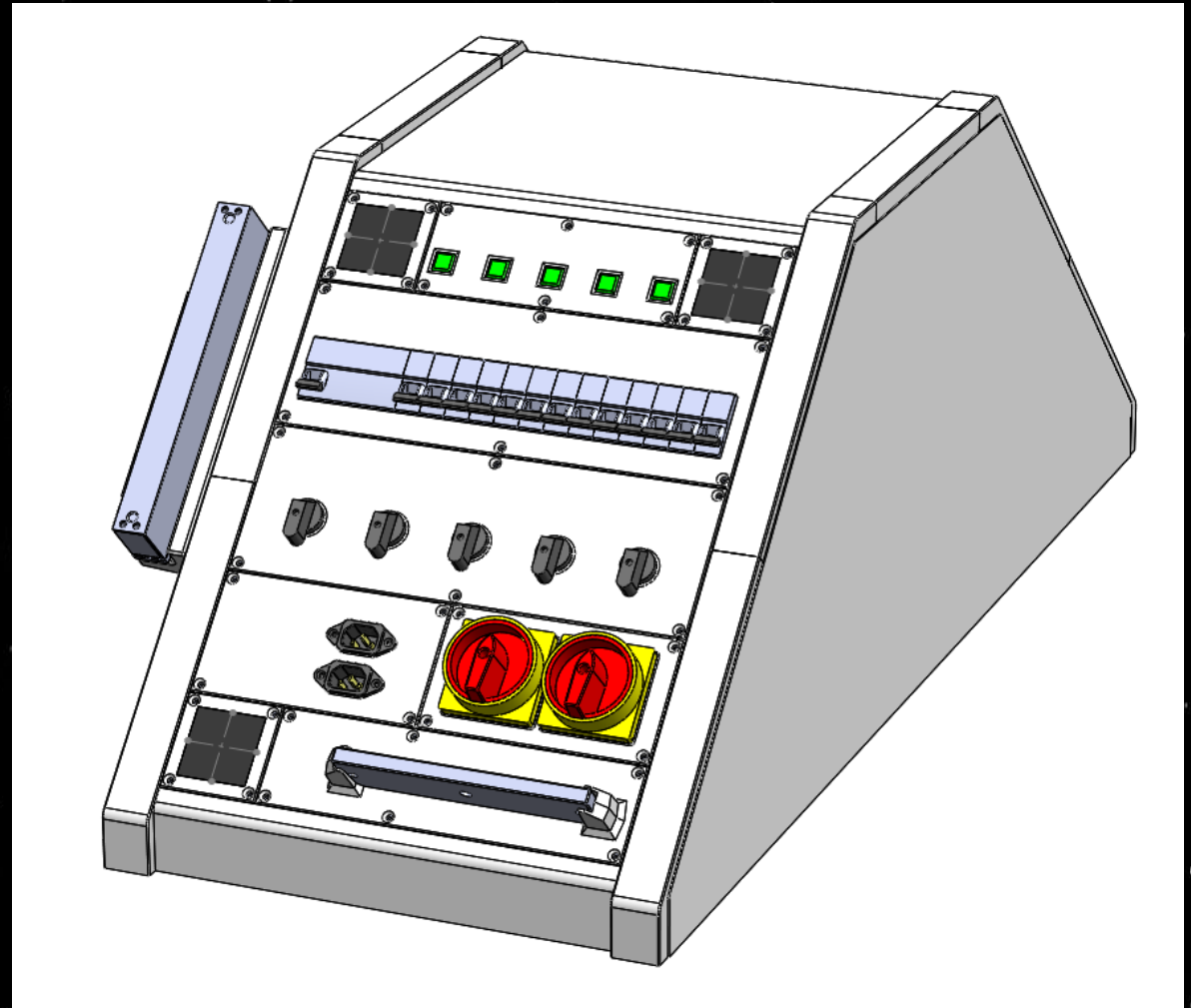
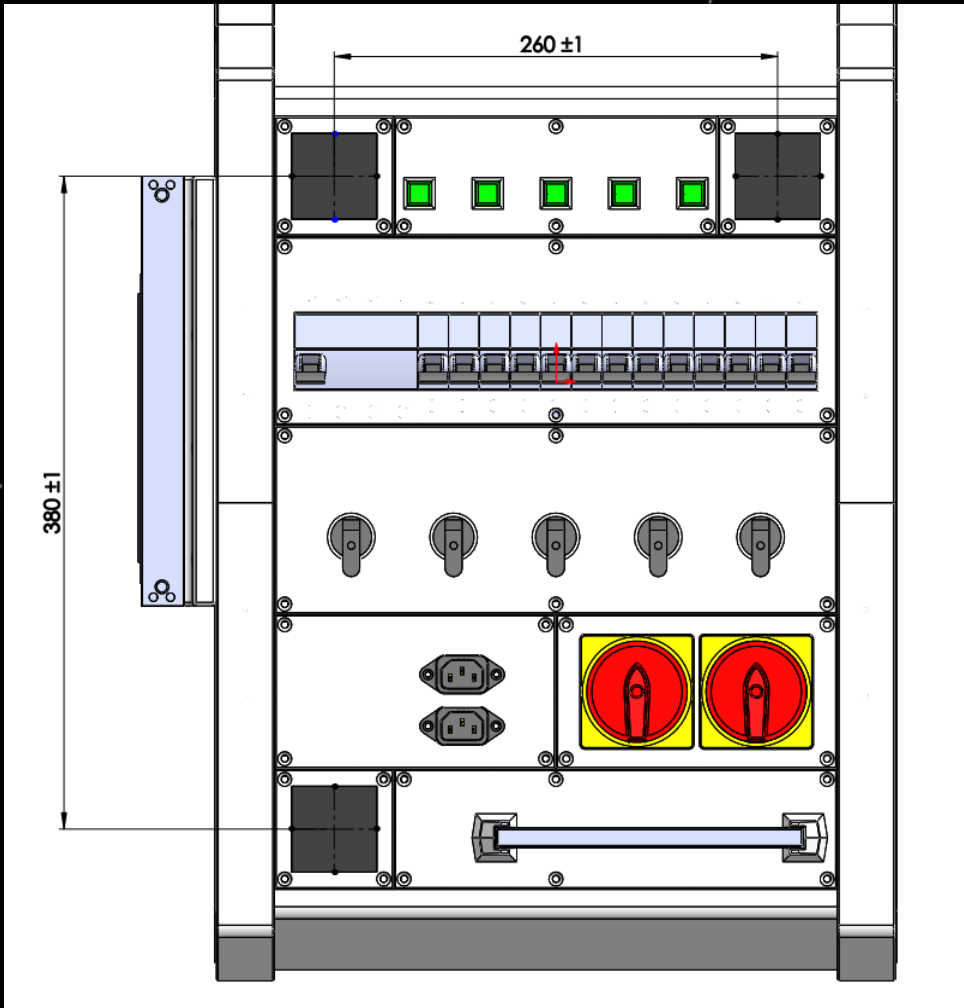
- Droning is to be performed in the drone cage (10x10x4m) that will be placed outside the MY.
- Drone will follow the markers from the ArUco 5x5 family, incl. the black envelope (the same as in case of the Traverse task, see above). Start and land points will use markers ID 101 and ID 102, in that order (**markers from the ORIGINAL ArUco LIBRARY shall be considered**). See <https://chev.me/arucogen/>.



Maintenance Task - Panel

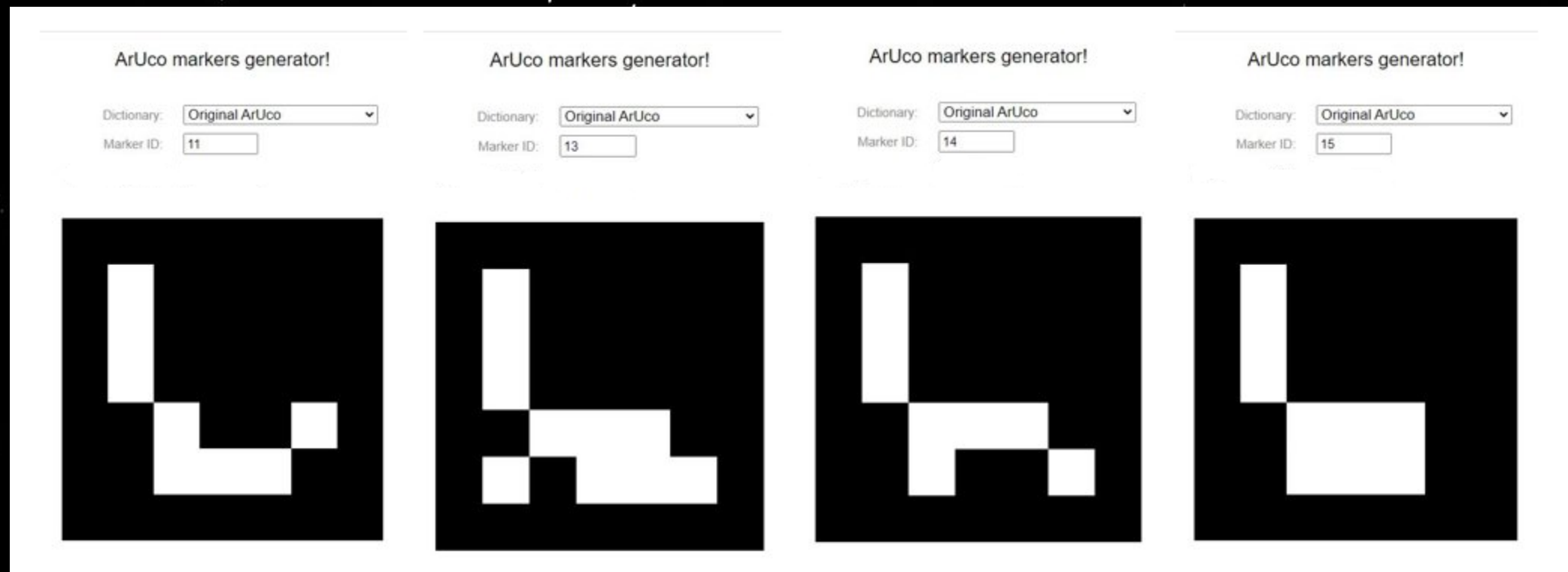


Maintenance Task - Panel



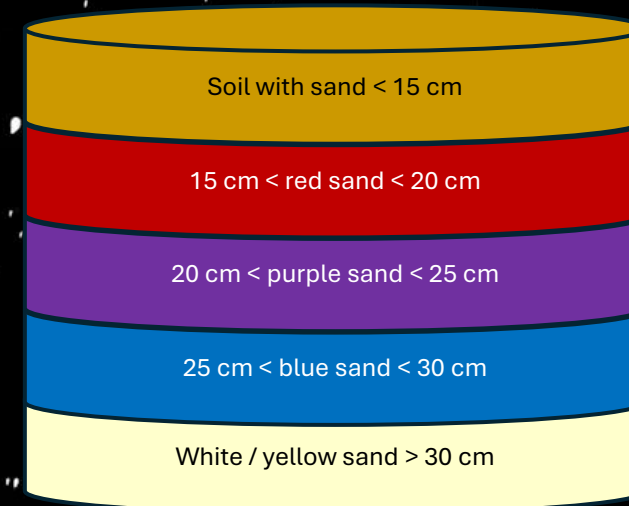
Maintenance Task - Markers

- Markers are indicated by the black squares. Actual size of the markers on the panel is 50x50mm.
- The following markers may be used on the panel: 11, 13, 14, 15 (**markers from the ORIGINAL ArUco LIBRARY shall be considered**). See <https://chev.me/arucogen/>.



Deep Sampling Sub-Task – Layers

- The following layers are foreseen, depending on the drilled-in depth (d):
 - $0\text{ cm} < d < 15\text{ cm}$
(Mars Yard ground mixed with sand)
 - $15\text{ cm} < d < 20\text{ cm}$ (red sand)
 - $20\text{ cm} < d < 25\text{ cm}$ (purple sand)
 - $25\text{ cm} < d < 30\text{ cm}$ (blue sand)
 - $d > 30\text{ cm}$ (white/yellow sand)



Tasks – Important Points



Remember to arrive at rover's weight / RF
check point in the team zone at least 30
minutes before your scheduled ride on the MY
!!!
This step has to be completed before your ride.

Remember to provide printed ERC 2026
[Traverse Planning Form](#) to judges (for
Navigation-Traverse Sub-Task). It is included in
the MY Update Report.



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See you in Kraków!

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